

**2110215 Programming Methodology
Final Project**

Group Name : TooOldToStudyProgMeth



**Group Members
6730084521 Chatrphol Ovanonchai**

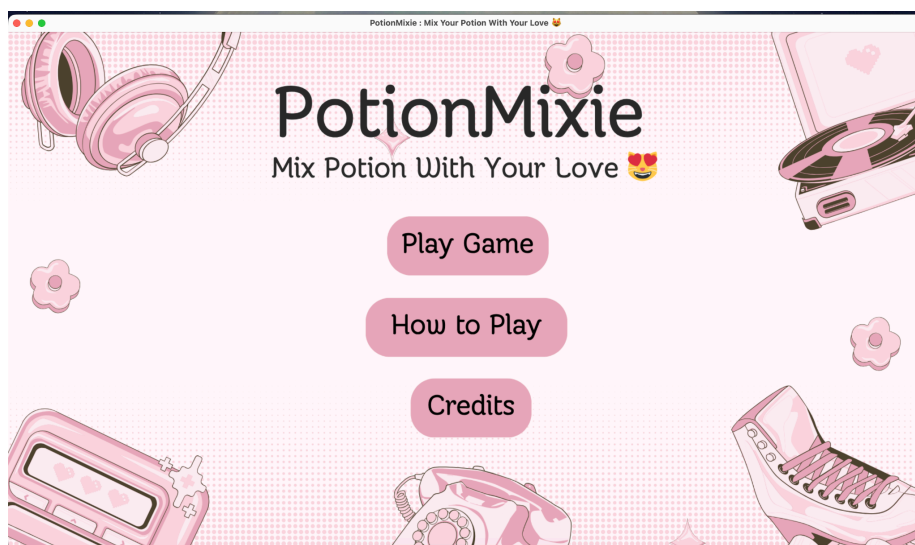
**This is a project report for course 2110215 Programming Methodology
Academic year 2025 Semester 2
Department of Computer Engineering , Chulalongkorn University**

Introduction

This game is inspired from Tower Defense Game & Power Puff Girls , in which players can destroy the enemies by mixing a potion. The objective of this game is to destroy all souls that appear in the conveyer belt like in Tower Defense Gameplay.

Overview of the game

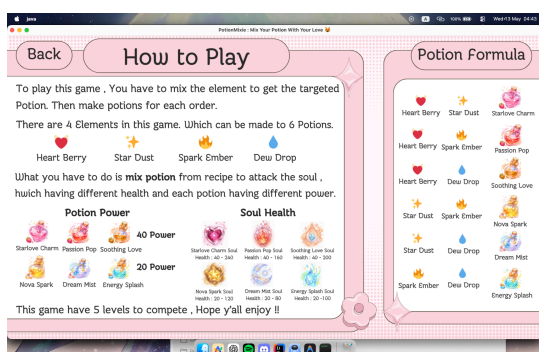
Main Menu Screen



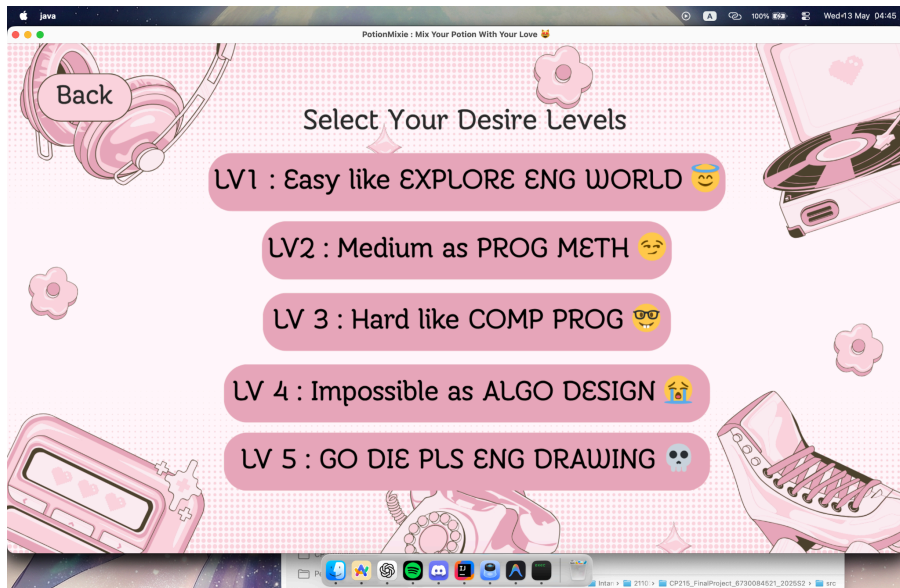
This screen is the main menu which every player in this game must see. Having 3 Buttons

1. Play Game Button : When pressed , will navigate to Level Selection Screen and the Game Screen
2. How To Play Button : When pressed , will giving details how to play this game and telling potion mixing formula
3. Credit Button : When pressed , represent myself and credits for this subject.

The picture below describe how **How to Play Screen & Credit Screen** be like



Level Selection Screen



This screen represents how players choose levels. Consists of 5 Levels of Difficulty.

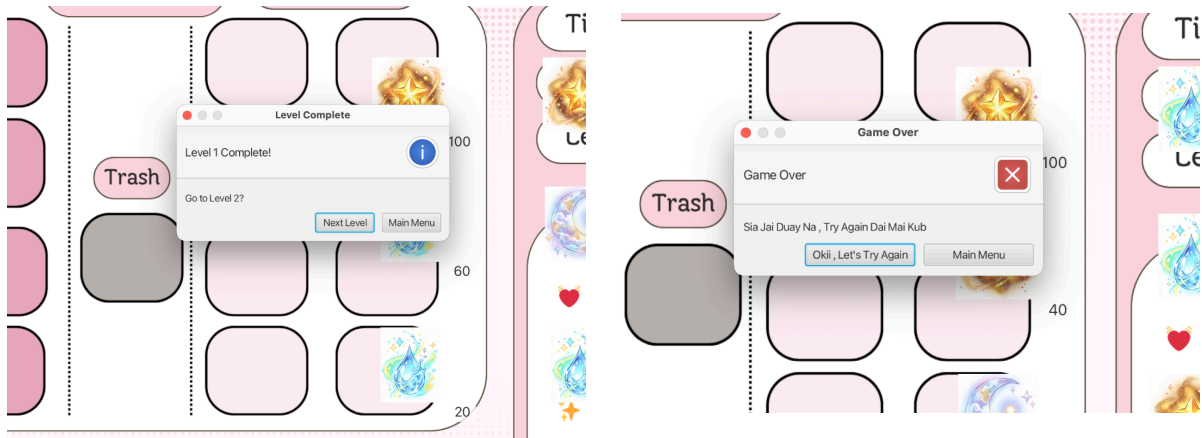
Game Screen



This screen represents the overall gameplay.

Note that I've got some problems about setting Layout , will fix it ASAP. (Now fixing it will get the game crashed)

Game Over Alert & Level Complete Alert



Note that I've got my program crashed after moving level , will fix it ASAP

This Alert is showing how game end (Game Over) or how game move to harder step (Level Complete)

Further Improvement

Needs to be fix

- Error layout from Merge Pane (which now changing it will crash program)
- Error when choosing next level or main menu from **Level Complete Alert**
- Image background which is not transparent
- Add more description about each function for JavaDoc

More features in the future

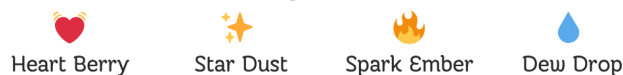
- More Levels
- Background Music
- Multiplayer mode

Gameplay

This game strategy is inspired from Tower Defense & Merge Game. What you have to do is mix elements into a potion then drag the potion to attack the soul.

Let's get into more details about this game elements

First , We've got 4 elements (which can't do anything except merging)



Then we can merge into 6 Potions from the formula and then potions have unique powers.



Third , we've souls which are like monsters to be attacked , but you have to use the same potion type to attack.

The formula of Health Power (HP) of each souls are derived below

- Starlove Charm Potion : $(\text{Random } 0-5) * 40 + 40 = [40 - 240 \text{ HP}]$
- Passion Pop Potion : $(\text{Random } 0-3) * 40 + 40 = [40 - 160 \text{ HP}]$
- Soothing Love Potion : $(\text{Random } 0-4) * 40 + 40 = [40 - 200 \text{ HP}]$
- Nova Spark Potion : $(\text{Random } 0-5) * 20 + 20 = [20 - 120 \text{ HP}]$
- Dream Mist Potion : $(\text{Random } 0-3) * 20 + 20 = [20 - 80 \text{ HP}]$
- Energy Splash Potion : $(\text{Random } 0-4) * 20 + 20 = [20 - 100 \text{ HP}]$

Next , We'll describe about how we design levels & orders according to levels when player pressed

Level 1 : Easy like EXPLORE ENG WORLD

- Having 5 souls to defeat
- 15 seconds to defeat each monster
- Bonus time for this level is 15 seconds
- So , the total time will be 90 Seconds

Level 2 : Medium as PROG METH

- Having 10 souls to defeat
- 15 seconds to defeat each monster
- Bonus time for this level is 30 seconds
- So , the total time will be 180 Seconds

Level 3 : Hard like COMP PROG

- Having 20 souls to defeat
- 12 seconds to defeat each monster
- No Bonus time for this level
- So , the total time will be 240 Seconds

Level 4 : Impossible as ALGO DESIGN

- Having 30 souls to defeat
- 10 seconds to defeat each monster
- No Bonus time for this level
- So , the total time will be 300 Seconds

Level 5 : GO DIE PLS ENG DRAWING

- Having **67** souls to defeat
- **6.7** seconds to defeat each monster
- Bonus time for this level is 2 seconds
- So , the total time will be 450 Seconds



These will be all of my gameplay. The next part will describe the UML diagram.

Class Diagram

The pictures below will describe the overall of the program.

Overall Diagram

> I can't put the image in here , so I'll insert the link instead. You can download and read them.

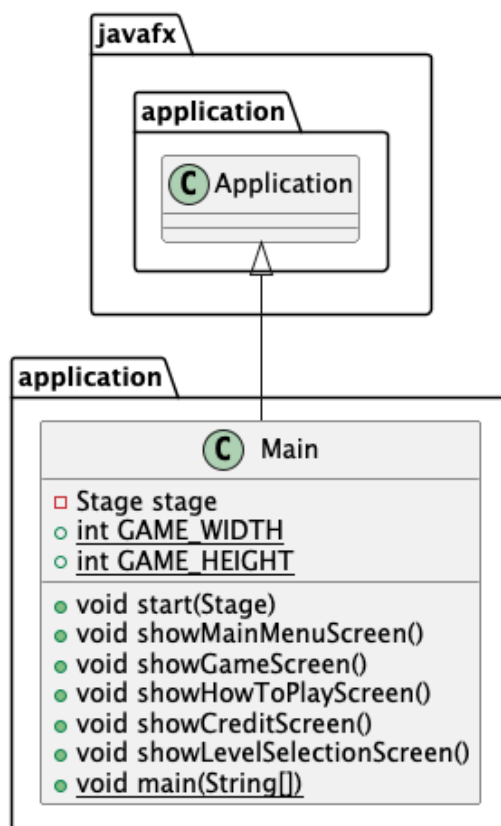
[CP215 Final Project UML All](#)

Diagram Elements

This program consists of 4 main package

- Package **application**
- Package **gameelement**
- Package **gamelogic**
- Package **gui**

First Part : Package **application** Diagram



Second Part : Package **gameelement** Diagram

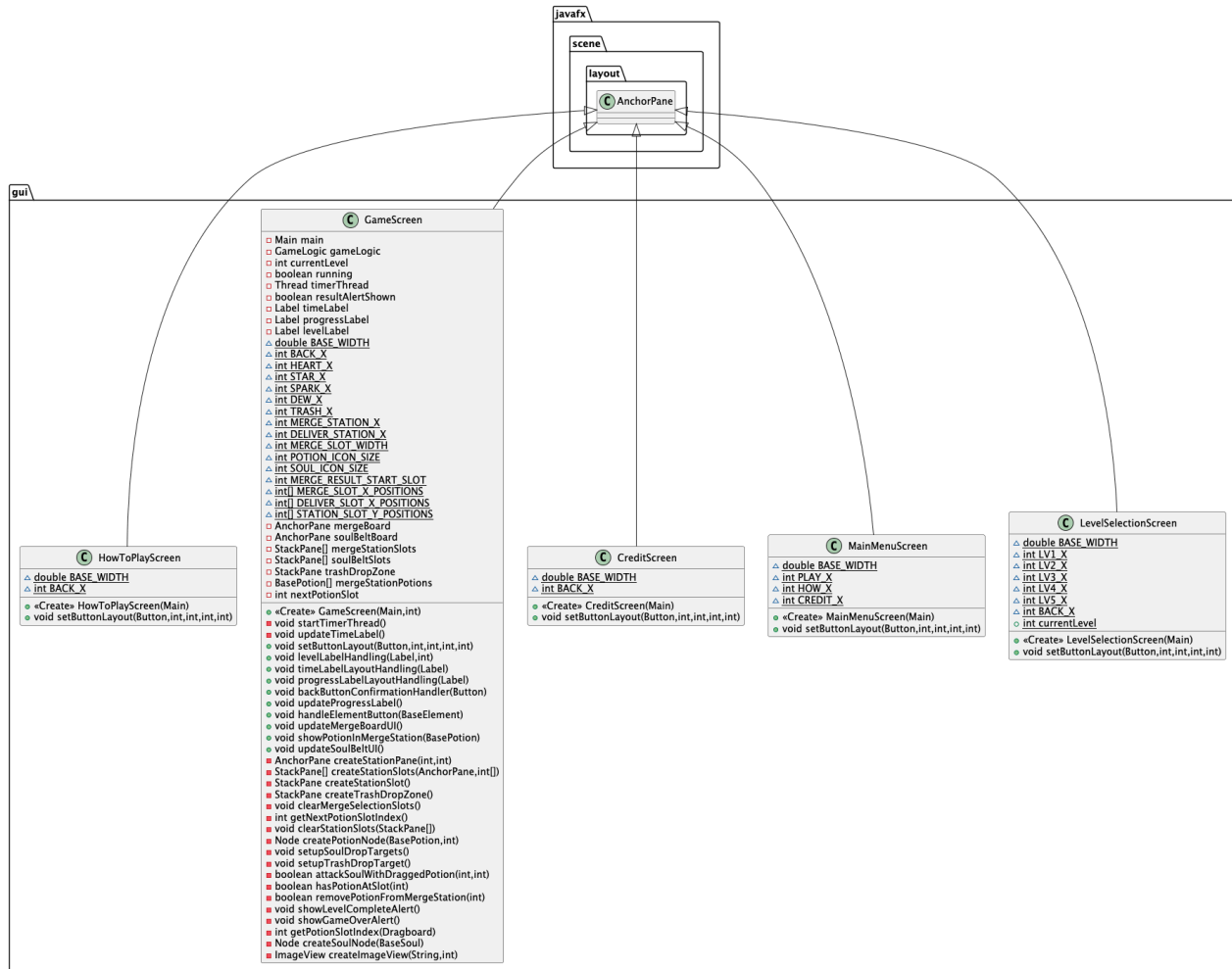
> I can't put the image in here , so I'll insert the link instead. You can download and read them.

[CP215 Final Project UML GameElement](#)

Third Part : Package **gamelogic** Diagram



Fourth Part : Package gui Diagram

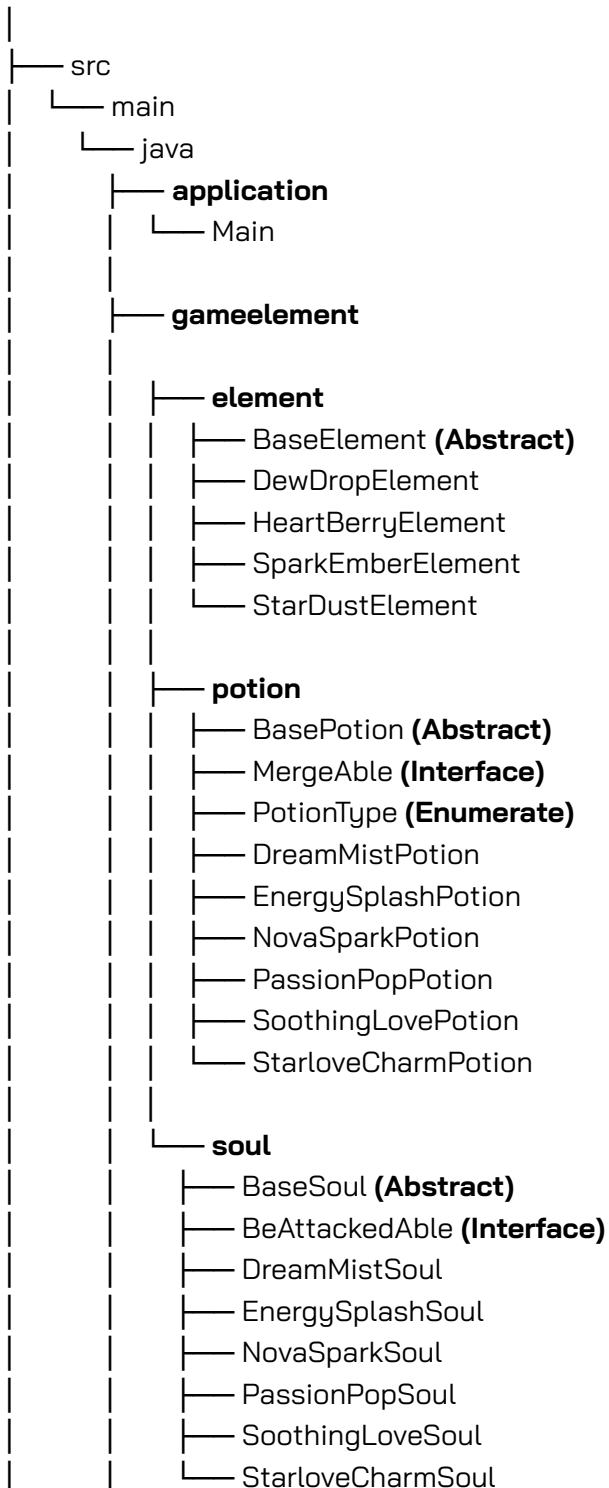


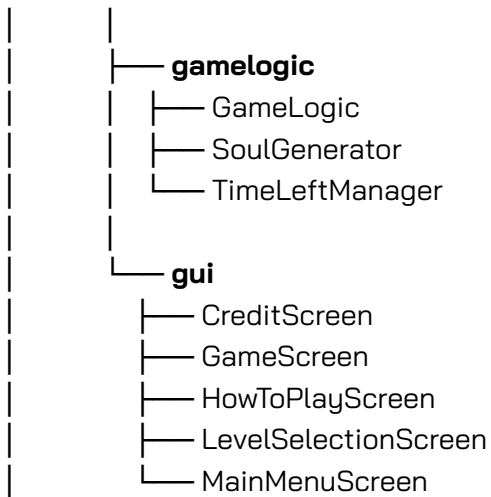
Class Description

The table below represent how each method done in each class

Structure of each package & class

CP215_FinalProject_6730084521_2025S2





Note that these symbols below represents for Access Modifiers

+ public , - private , # protected , static

Java Documentation

<https://2110215-final-project-potion-mixie.vercel.app/>